

SMART RESUME ANALYZER

Abstract

The Smart Resume Analyzer is an AI-powered application designed to evaluate resumes and provide actionable feedback for job seekers. The system analyses resumes uploaded in PDF format and performs resume quality assessment, ATS (Applicant Tracking System) compatibility analysis, skill gap detection, career readiness evaluation, and AI-powered resume review.

The application extracts text from resumes and compares candidate skills against role-specific requirements for various job profiles. ATS compatibility is calculated through keyword matching, while a custom scoring system evaluates resume completeness and quality. Google Gemini is integrated to generate detailed AI-powered resume assessments and recommendations.

An interactive Streamlit dashboard was developed to provide real-time analysis, visualisations, skill matching reports, career roadmaps, and downloadable PDF reports. The project demonstrates the practical application of Generative AI and Natural Language Processing techniques in career development and recruitment support.

1. Introduction

Recruitment processes have increasingly adopted Applicant Tracking Systems (ATS) to screen and filter resumes before they reach hiring managers. Many qualified candidates are rejected due to poor resume structure, missing keywords, or inadequate presentation of skills and achievements.

Traditional resume evaluation methods are often subjective and time-consuming. Artificial Intelligence provides an effective solution by automating resume analysis and generating personalized recommendations.

The Smart Resume Analyzer was developed to help candidates understand their resume strengths and weaknesses, improve ATS compatibility, identify missing skills, and receive AI-generated feedback through an easy-to-use web application.

2. Objectives

The main objectives of this project are:

- Analyze resumes uploaded in PDF format.
- Calculate resume quality and completeness scores.

- Evaluate ATS compatibility for selected job roles.
 - Identify matched and missing skills.
 - Generate AI-powered resume analysis using Google Gemini.
 - Provide personalized recommendations and career roadmaps.
 - Generate downloadable PDF reports.
 - Develop an interactive dashboard for real-time analysis.
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3. System Description

The Smart Resume Analyzer consists of multiple modules that work together to evaluate resumes.

Modules Used

Resume Parser

Extracts text content from uploaded PDF resumes.

Resume Score Calculator

Evaluates the overall quality and completeness of the resume.

ATS Compatibility Checker

Compares resume content with role-specific skill requirements.

Skill Gap Analyzer

Identifies missing skills required for the selected target role.

Feedback Generator

Provides recommendations for resume improvement.

Career Roadmap Generator

Suggests learning paths and development activities based on career goals.

AI Resume Analyzer

Uses Google Gemini to generate detailed professional feedback.

PDF Report Generator

Creates downloadable resume assessment reports.

4. Methodology

Resume Processing

The system follows the following workflow:

Resume Upload



PDF Text Extraction



Resume Scoring



ATS Keyword Matching



Skill Gap Analysis



Feedback Generation



AI Resume Review



Dashboard Visualization & PDF Report

ATS Matching Process

The ATS compatibility score is calculated by comparing the skills found in the resume with predefined role-specific skills.

Matched skills contribute positively to the score while missing skills are highlighted for improvement.

Resume Evaluation

The resume is evaluated based on:

- Presence of projects
- Certifications
- Internship experience
- LinkedIn profile
- GitHub profile (for technical roles)
- Measurable achievements
- Resume completeness

AI Analysis

Google Gemini is used to perform advanced resume analysis and generate personalized recommendations based on resume content.

5. Technologies Used

TECHNOLOGY	PURPOSE
Python	Core Programming Language
Streamlit	Dashboard Development
Google Gemini	AI Resume Analysis
Pypdf2	PDF Text Extraction
Plotly	Data Visualization
Reportlab	PDF Report Generation
Github	Version Control
Streamlit Cloud	Deployment

6. Results and Features

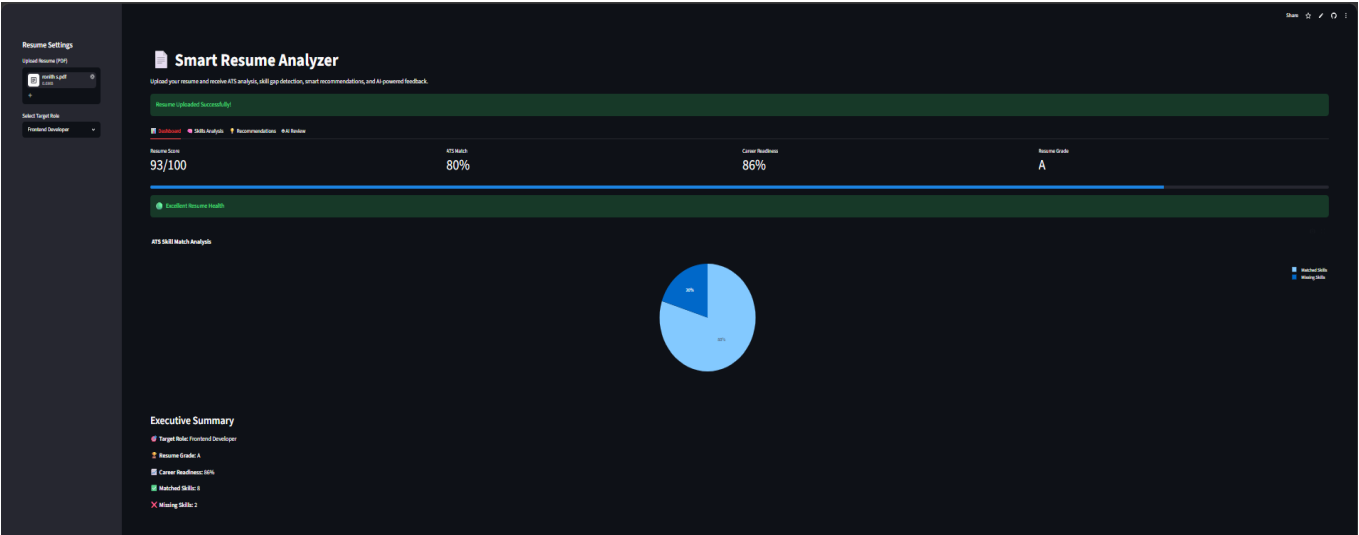
The Smart Resume Analyzer successfully provides:

- Resume Score Evaluation
- ATS Compatibility Assessment
- Career Readiness Score
- Resume Grade Calculation
- Skill Match Analysis
- Missing Skill Identification
- Career Roadmap Recommendations
- AI-Powered Resume Review
- Downloadable PDF Reports

Dashboard Features

Users can:

- Upload PDF resumes.
- Select a target job role.
- View ATS compatibility analysis.
- View skill match visualizations.
- Receive personalized recommendations.
- Generate AI-powered resume reviews.
- Download detailed PDF reports.



DashboardSkills AnalysisRecommendationsAI Review

Career Roadmap

Build Portfolio Website

Build E-Commerce Frontend

Build Task Management App

Master React

Resume Improvements

Add your LinkedIn profile.

Skill Development

Learn and add: REST API, Frontend Development

Experience Building

Excellent! Your resume already includes projects and practical experience.

Certifications

Add relevant certifications.

DashboardSkills AnalysisRecommendationsAI Review

Generate AI Resume Analysis

Here is an analysis of your resume:

1. Resume Strength Summary

The resume presents a highly experienced Java/J2EE Developer with over 9 years in the Software Development Life Cycle (SDLC). A significant strength is the broad and deep technical skillset spanning core Java, J2EE technologies, modern front-end frameworks (AngularJS, ReactJS, Node.js), robust back-end frameworks (Spring, Hibernate, Struts), web services (RESTful, SOAP), various databases (Oracle, MySQL, MongoDB), and cloud platforms (AWS). The candidate demonstrates practical experience with Agile methodologies, Test-Driven Development (TDD), and Continuous Integration (Jenkins). The inclusion of specific tools like FORTIFY for security and DynaTrace/Splunk for monitoring, along with experience in crucial phases like requirements gathering, design, development, testing, and maintenance, highlights a well-rounded and capable senior developer profile. Expertise in design patterns and object-oriented principles further strengthens the technical foundation.

2. Technical Skills Identified

- Languages: Java (Core Java, J2EE, JDK 1.7, 1.8), C++, HTML, XML, CSS, JavaScript, SQL, PL/SQL
- Java Technologies: Core Java, Multithreading, Servlets, JSP, JSTL, Java Beans, Applets, JDBC, EJB (Session Beans, Message Driven Beans)
- Frameworks/Libraries: Spring (MVC, Boot, Cloud, DI, IOC, Remoting, ORM, DAO, AOP, JMS, Business Integration), Hibernate, Struts (1.2, Tiles, Validator), JSF, PrimeFaces, iBatis, AJAX, JQuery, Bootstrap, Curam Framework, Log4j, JUnit, Mockito, Jasmine, Karma, Sencha 4.5, GWT (Google Web Toolkit), Dojo
- Web Services: SOAP (JAX-WS), REST (JAX-RS), Microservices
- Cloud Platforms: AWS (API's, EC2, RDS, Lambda), Spring Cloud
- Databases: Oracle (9i/10g/11g), MySQL, Sybase, DB2, MS SQL Server, MongoDB
- ORM Tools: Hibernate, HQL, iBatis, JPA
- Web/Application Servers: Apache Tomcat, JBoss, Oracle WebLogic Server, Glassfish Application Server, IBM WebSphere Application Server (8.5), Node.js (as server/runtime)
- Build/CI/CD Tools: Maven, Ant, Jenkins, Grunt, Gulp
- Version Control: SVN, CVS, GIT, Visual Source Safe
- Operating Systems: Windows (XP/Vista/7), UNIX, Linux
- IDEs/Tools: Eclipse, Rational Application Developer (RAD), WebSphere Studio Application Developer 5.0, Rational Software Architect (RSA), STS, IntelliJ, Clion, Brackets, SQL Developer, Putty, Splunk, HP ALM QC 11.0, JIRA, DynaTrace, FORTIFY, SOAPUI, TOAD
- Front-end: HTML5, CSS3, JavaScript, JQuery, Angular JS (1.0, 2.0), React JS, Node.js, Backbone.js, DHTML, Custom Tag Libraries
- Messaging: JMS, RabbitMQ
- Design/Methodologies: SDLC, OOA, OOD, OOP, UML, MVC, DAO, Factory pattern, Session Facade, Singleton, Business Delegate, DTO, Front Controller, Rational Unified Process (RUP), Agile, Scrum, TDD, BDD, Waterfall (mentioned in older project context)
- Scripting: UNIX Shell scripts, Linux shell scripts
- Security: SQL Injection avoidance, FORTIFY tool

3. Missing Skills

While the resume is rich in technologies, to align with the cutting edge of modern Java/Cloud development and senior roles, the following areas could be strengthened or explicitly mentioned:

- Containerization & Orchestration: Deeper experience with Docker and Kubernetes for microservices deployment and management.
- Advanced CI/CD: More detailed mention of advanced Jenkins pipelines (declarative/scripted), artifact management (e.g., Nexus, Artifactory), and deployment strategies (e.g., Blue/Green, Canary releases).
- Infrastructure as Code (IaC): Tools like Terraform or AWS CloudFormation for automated infrastructure provisioning.

7. Recommended Certifications

1. Oracle Certified Professional (OCP): Java SE Developer

2. Spring Certified Professional: This certification from

3. AWS Certified Developer - Associate / AWS Certified

4. Certified Kubernetes Application Developer (CKAD):

5. Professional Scrum Master (PSM I) / Professional Scrum

Download PDF Report

SMART RESUME ANALYZER

Candidate Assessment Report

Generated On: 06-06-2026

Target Role: Frontend Developer

Resume Score: 93/100

ATS Compatibility: 80%

Career Readiness: 86%

Resume Grade: A

Matched Skills

• HTML

• CSS

• JavaScript

• React

• Bootstrap

• Responsive Design

• UI

• Git

Missing Skills

• REST API

7. Challenges and Improvements

Several challenges were encountered during development.

Resume Parsing Challenges

Different resume formats required robust text extraction and preprocessing.

ATS Matching Accuracy

Skill matching was improved through the use of aliases and keyword normalization.

AI Integration

Prompt engineering techniques were used to improve the quality and relevance of Gemini-generated feedback.

Deployment Challenges

Application deployment required dependency management, GitHub integration, and secure API key configuration using Streamlit Secrets.

These improvements contributed to a more reliable and user-friendly application.

8. System Implementation

A web-based dashboard was developed using Streamlit.

The dashboard includes:

- Dashboard Tab
- Skills Analysis Tab
- Recommendations Tab
- AI Review Tab

Users can interactively analyze resumes and receive real-time feedback.

The system is deployed online using Streamlit Cloud and integrated with GitHub for continuous updates.

9. Conclusion and Future Scope

Conclusion

The Smart Resume Analyzer successfully combines Artificial Intelligence, ATS evaluation techniques, and interactive data visualization to provide comprehensive resume analysis.

The application helps users improve resume quality, increase ATS compatibility, identify skill gaps, and receive personalized career recommendations. Integration of Google Gemini enhances the quality of feedback and makes the system more intelligent and user-centric.

Future Scope

- Job Description Matching
 - Resume Ranking System
 - Multi-Language Resume Support
 - Interview Question Generation
 - Resume Optimization Suggestions
 - Integration with Job Portals
 - Real-Time Job Recommendations
 - Advanced ATS Simulation
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References

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2. Google Gemini API Documentation
3. Plotly Documentation
4. ReportLab Documentation
5. PyPDF2 Documentation
6. Applicant Tracking System(ATS) Research Articles
7. Natural Language Processing (NLP) Documentation